Day 1 Engineering Report

Develop Backend & Database Integration — PAGE International Platform

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Date: October 28, 2026

100% COMPLETED

Day 1 Focus & Objective: Designed and deployed the foundational PostgreSQL 16 relational data architecture, implemented schema constraints and automated database triggers, built custom CLI migration and health monitoring tooling, seeded the baseline commercial catalog, and migrated the entire schema to production **Supabase Cloud**.

16	28	2	100%
Relational Tables	Catalog Products Seeded	DevOps CLI Utilities	Supabase Cloud Sync

DAY 1 MILESTONES & COMPLETED SUBTASKS

1. Relational Database Schema Design (16 Tables)

Verified

Architected normalized PostgreSQL schema supporting multi-table relationships with foreign keys, ON DELETE rules, and check constraints. Formatted technical specifications as queryable JSONB arrays.

`products` `categories` `brands` `branches` `quotes` `orders` `customers` `audit_logs`

2. Automated PostgreSQL Triggers & Business Rules

Verified

Implemented automated stored procedures for timestamp updating (`set_updated_at`), dynamic order total recalculations (`calculate_order_totals`), and automated administrative audit trail logging.

`PL/pgSQL` `Audit Triggers` `Auto-Timestamps`

3. DevOps CLI Migration & Health Monitoring Runner

Verified

Created standalone CLI utilities for database operations. Built an idempotent migration runner tracked by `schema_migrations` and a live health check status reporter.

`npm run db:migrate` `npm run db:status` `schema_migrations`

4. Enterprise Commercial Catalog & Seed Engine

Verified

Developed automated seed pipeline loading 28 commercial machines (2-post lifts, wheel aligners, diagnostic tools), 6 categories, 10 international brands, 3 regional service branches, and 5 role-based admin users.

`npm run db:seed` `28 Products` `5 Admins` `3 Branches`

5. Production Supabase Cloud Migration

Verified

Prepared and executed consolidated cloud migration scripts against Supabase Cloud database. Resolved dynamic database resolution and cloud payload constraints. All 16 tables and catalog seeds deployed with 0 errors.

`Supabase Cloud` `PostgreSQL 16` `Production Live`

DATABASE ENTITIES VERIFICATION TABLE

ENTITY / TABLE	RECORD COUNT	PRIMARY KEY / IDENTIFIER	STATUS
products	28 records	`id (UUID) / slug (TEXT UNIQUE)`	Live in Cloud
categories & brands	6 categories, 10 brands	`id (UUID) / slug (TEXT UNIQUE)`	Live in Cloud
branches & admins	3 branches, 5 admins	`id (UUID) / email (TEXT UNIQUE)`	Live in Cloud
customer & commerce tables	quotes, orders, bookings, customers	`id (UUID) + RLS security policies`	Live in Cloud